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Gohar Irfan Chaudhry

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Research Interests

My research goal is to design and develop resource-efficient systems for serving AI workloads at scale. I work on various aspects of the cloud platform, including runtimes and operating systems to hardware/software codesign, towards realizing this goal.

Education

- **Massachusetts Institute of Technology** - *PhD Computer Science* (Sept 2022 - Present)
  - Advisor: Prof. Adam Belay
- **University of Illinois at Urbana-Champaign** - *MSc Computer Science* (August 2016 - May 2018)
  - Thesis: Network Analysis and Verification
- **Lahore University of Management Sciences** - *BSc Computer Science* (August 2012 - May 2016)

Employment

Research Software Engineer **Microsoft Research** (Oct 2019 - Aug 2022)

Systems Research Group

- Working on improving efficiency of cloud platforms, in particular serverless infrastructure.

Software Engineer **Microsoft** (July 2018 - Oct 2019)

Windows Operating System Group

- Implementation of a virtualized audio stack in the OS to run cross-platform on PC and Xbox.

Publications

- **Murakkab: Resource-Efficient Agentic Workflow Orchestration in Cloud Platforms**
  G I Chaudhry, E Choukse, H Qiu, I Goiri, R Fonseca, A Belay, R Bianchini
  (OSDI 2026) USENIX Symposium on Operating Systems Design and Implementation
- **Unleashing The Potential of Datacenter SSDs by Taming Performance Variability**
  G I Chaudhry, A Bhardwaj, Z Ruan, A Belay
  (NSDI 2026) USENIX Symposium on Networked Systems Design and Implementation
- **Towards Resource-Efficient Compound AI Systems**
  G I Chaudhry, E Choukse, I Goiri, R Fonseca, A Belay, R Bianchini
  (HotOS 2025) ACM SIGOPS Workshop on Hot Topics in Operating Systems
- **Architecture-Level Modeling of Photonic Deep Neural Network Accelerators**
  T Andrulis, G I Chaudhry, V M. Suriyakumar, J S. Emer, V Sze
  (ISPASS 2024 - Poster Session) IEEE International Symposium on Performance Analysis of Systems and Software
- **Making Kernel Bypass Practical for the Cloud with Junction**
  J Fried, G I Chaudhry, E Saurez, E Choukse, Í Goiri, S Elnikety, R Fonseca, A Belay
  (NSDI 2024) USENIX Symposium on Networked Systems Design and Implementation
- **Palette Load Balancing: Locality Hints for Serverless Functions**
  M Abdi, S Ginzburg, C Lin, J M Faleiro, Í Goiri, G I Chaudhry, R Bianchini, D S Berger, R Fonseca
  (EuroSys 2023) European Conference On Computer Systems
- **Memory-Harvesting VMs in Cloud Platforms**
  A Fuerst, S Novakovic, Í Goiri, G I Chaudhry, P Sharma, K Arya, K Broas, E Bak, M Iyigun, R Bianchini
  (ACM ASPLOS 2022) Conference on Architectural Support for Programming Languages and Operating Systems
- **Faa$T: A Transparent Auto-Scaling Cache for Serverless Applications**
  F Romero, G I Chaudhry, Í Goiri, P Gopa, P Batum, N J. Yadwadkar, R Fonseca, C Kozyrakis, R Bianchini
  (ACM SoCC 2021) Symposium on Cloud Computing

- **Faster and Cheaper Serverless Computing on Harvested Resources**
  Y Zhang, Í Goiri, G I Chaudhry, R Fonseca, S Elnikety, C Delimitrou, R Bianchini
  (ACM SOSP 2021) Symposium on Operating Systems Principles
- **Serverless in the Wild: Characterizing and Optimizing the Serverless Workload at a Large Cloud Provider**
  M Shahrad, R Fonseca, Í Goiri, G I Chaudhry, P Batum, J Cooke, E Laureano, C Tresness, M Russinovich, R Bianchini
  (USENIX ATC 2020) Annual Technical Conference - Winner of the Community Award
- **High Coverage Testing of Softwarized Networks**
  S Prabhu, G I Chaudhry, B Godfrey and M Caesar
  (ACM SIGCOMM 2018 - SecSoN) Workshop on Security in Softwarized Networks
- **MegaVM - A Memory Enhancing Framework for Datacenters**
  R Tahir, G I Chaudhry, B Bakht, H Sharif, F Zaffar, M Caesar
  (USENIX NSDI 2016 - Poster Session) Symposium on Networked Systems Design and Implementation

Teaching Assistant

- **University of Illinois at Urbana-Champaign** (2017 - 2018)
  CS484 Parallel Programming *(OpenMP, MPI, Charm++)*, *CS125* Intro to Programming *(Java)*
- **Lahore University of Management Sciences** (2013 - 2016)
  CS473 Network Security *(C, C++)*, *CS382* Network-Centric Computing *(Java)*, *CS200* Intro to Programming *(C++)*

Other Experiences

- **Recruiting** (2019 - 2022)
  - Conducting technical interviews for Microsoft